BIO503 BIOPHYSICS MIDTERM MEGA FILE

ALL DATA FROM PAST PAPERS (2018-2022)

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Past papers

Question no 1: What is uniqueness in Biophysics?

Answer:

1. Biophysics is a growing enterprise world-wide, driven primarily by the widespread realization of the major contributions made to biological science by a combination of truly state-of-the-art physical measurements with modern molecular biology.
2. The field occupies a unique and central position at the intersection of the biological, chemical, physical, and computational sciences
3. Development and advancement of new techniques and approaches.
4. Biophysicists as a group most often develop the novel, sophisticated experimental methods that reveal molecular level details with unique clarity.

Question no 2: Difference between intra-molecular and intermolecular forces of attraction?

Answer:

- **Intramolecular:**

Intramolecular forces are the forces that hold atoms together within a molecule.

- **Intermolecular:**

Intermolecular forces are forces that exist between molecules. These are also known as **van der Waals forces**, named after Johannes van der Waals who first postulated them. Intermolecular

forces are typically weaker than intramolecular forces, and account for the bulk properties of matter (e.g., boiling point, melting point, etc.)

Question no 3: Define Ion-induced dipole forces and explain it?

Answer: An ion-induced dipole attraction is a weak attraction that results when the approach of an ion induces a dipole in an atom or in a nonpolar molecule by disturbing the arrangement of electrons in the nonpolar species.

Explanation:

The force of attraction between a non-polar molecule and an ion that may lie in its vicinity is called ion-induced dipole force. The strength of this induced dipole depends on how easily the electron cloud can be distorted, i.e., the bigger the molecule, the stronger is the dipole induced.

Example: Hemoglobin is a protein found in the red blood cells, and its function is to carry oxygenated blood to various parts of the body.

Question no 4: What are Intramolecular hydrogen bonds?

Answer: Intramolecular hydrogen bonds are those which occur within one single molecule. This occurs when two functional groups of a molecule can form hydrogen bonds with each other. In order for this to happen, both a hydrogen donor a hydrogen acceptor must be present within one molecule, and they must be within close proximity of each other in the molecule.

For example, intramolecular hydrogen bonding occurs in ethylene glycol ($C_2H_4(OH)_2$) between its two hydroxyl groups due to the molecular geometry.

Question no 5: What are intermolecular hydrogen bonds?

Answer: Intermolecular hydrogen bonds occur between separate molecules in a substance. They can occur between any number of like or unlike molecules as long as hydrogen donors and acceptors are present in positions where they can interact with one another.

For example, intermolecular hydrogen bonds can occur between NH_3 molecules alone, between H_2O molecules alone, or between NH_3 and H_2O molecules.

Question no 6: Write the importance of hydrogen bonds?

Answer:

- Source of unique properties of water
- Structure and function of proteins
- Structure and function of DNA
- Structure and function of polysaccharides
- Binding of a substrate to enzymes
- Binding of hormones to receptors

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Question no 7: Predict which will have the higher boiling point: N2 or CO. Explain your reasoning?

Answer: CO and N2 are both diatomic molecules with masses of about 28 amu, so they experience similar London dispersion forces. Because CO is a polar molecule, it experiences dipole-dipole attractions.

Because N2 is nonpolar, its molecules cannot exhibit dipole-dipole attractions. The dipole-dipole attractions between CO molecules are comparably stronger than the dispersion forces between nonpolar N2 molecules, so CO is expected to have the higher boiling point.

Question no 8: Write and define types of RNA?

Answer:

Three main types of RNA are involved in protein synthesis. They are messenger RNA (mRNA), transfer RNA (tRNA), and ribosomal RNA (rRNA)

Messenger RNA (mRNA): In protein synthesis, mRNA carries genetic codes from the DNA in the nucleus to ribosomes, the sites of protein translation in the cytoplasm.

Ribosomal RNA (rRNA): The ribosome protein subunits are encoded by rRNA and are synthesized in the nucleolus.

Transfer RNA (tRNA): It has a long-established role in protein synthesis. The tRNA molecule serves as an adaptor between the genetic instructions written in nucleic acid sequences and the protein products encoded in genes.

Question no 9: What are group II and Polar unchanged amino acids?

Answer: Group II amino acids are serine, cysteine, threonine, tyrosine, asparagine, and glutamine. The side chains in this group possess a spectrum of functional groups. However, most have at least one atom (nitrogen, oxygen, or sulfur) with electron pairs available for hydrogen bonding to water and other molecules.

Question no 10: Define any five terms from these: Thermal contact, Thermal equilibrium, Thermal expansion, Conduction, Convection, Radiation, Insulation.

Answer:

Thermal contact: When two solid bodies are in contact, a resistance to heat transfer exists between the bodies.

Thermal equilibrium: A state of a system in which all parts are at the same temperature.

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Thermal expansion: Thermal expansion is the tendency of matter to change its shape, area, volume, and density in response to a change in temperature, usually not including phase transitions.

Conduction: Conduction is the process by which heat energy is transmitted through collisions between neighboring atoms or molecules.

Convection: process by which heat is transferred by movement of a heated fluid such as air or water.

Radiation: Radiation is energy that comes from a source and travels through space at the speed of light.

Insulation: material that is used to stop the passage of electricity, heat, or sound from one conductor to another.

Question no 11: Define standard-state enthalpy of reaction and Enthalpy of formation?

Answer:

Standard enthalpy of reaction: The Heat of Reaction (also known and Enthalpy of Reaction) is the change in the enthalpy of a chemical reaction that occurs at a constant pressure. It is a thermodynamic unit of measurement useful for calculating the amount of energy per mole either released or produced in a reaction.

Standard enthalpy of formation: The standard enthalpy of formation is a measure of the energy released or consumed when one mole of a substance is created under standard conditions from its pure elements. The symbol of the standard enthalpy of formation is ΔH f. = A degree signifies that it's a standard enthalpy change.

Question no 12: Define Lattice enthalpy?

Answer: The amount of energy required to completely separate one mole of the solid ionic compound into constituent gaseous ions is known as lattice enthalpy. Lattice enthalpy is a measure of the strength of an ionic compound.

The lattice enthalpy of a crystalline solid is a measure of the energy released when the ions are combined to make this compound. The lattice enthalpy is indirectly determined by the use of the Born-Haber Cycle. This procedure is based on Hess's law.

Question no 13: Define entropy and write its units?

Answer: The measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of entropy is also a measure of the molecular disorder, or randomness, of a system.

The units of entropy are J/K. The temperature in this equation must be measured on the absolute, or Kelvin temperature scale.

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Question no 14: Define first law of thermodynamics?

Answer: The first law of thermodynamics, or the law of conservation of energy. The change in a system's internal energy is equal to the difference between heat added to the system from its surroundings and work done by the system on its surroundings.

Question no 15: Define and explain Second law of thermodynamics?

Answer: The second law of thermodynamics. Heat does not flow spontaneously from a colder region to a hotter region, or, equivalently, heat at a given temperature cannot be converted entirely into work.

One simple statement of the law is that heat always moves from hotter objects to colder objects (or "downhill"), unless energy is supplied to reverse the direction of heat flow. Another definition is: "Not all heat energy can be converted into work in a cyclic process

Question no 16: Define zeroth law of thermodynamics?

Answer: The zeroth law of thermodynamics. When two systems are each in thermal equilibrium with a third system, the first two systems are in thermal equilibrium with each other. This property makes it meaningful to use thermometers as the "third system" and to define a temperature scale.

Question no 17: What do you know about Fahrenheit, Celsius and kelvin?

Answer:

Fahrenheit: relating to or having a temperature scale on which the boiling point of water is at 212 degrees above the zero of the scale and the freezing point is at 32 degrees above zero.

Celsius: a measurement of temperature on a standard in which 0° is the temperature at which water freezes, and 100° the temperature at which it boils

Kelvin: The kelvin (abbreviation K) is the Standard International (SI) unit of thermodynamic temperature. One kelvin is formally defined as 1/273.16) of the thermodynamic temperature of the triple point of pure water (H_2O).

Question no 18: Define metabolic pathway?

Answer: A metabolic pathway is a linked series of chemical reactions occurring within a cell. The reactants, products, and intermediates of an enzymatic reaction are known as metabolites, which are modified by a sequence of chemical reactions catalyzed by enzymes.

Question no 19: Difference between passive and active transport?

Answer: Key Points

- Active transport requires energy for the movement of molecules whereas passive transport does not require energy for the movement of molecules.

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- In active transport, the molecules move against the concentration gradient whereas in passive transport, the molecules move along the concentration gradient.
- Uptake of glucose in the human intestine works on the principle of active transport.
- Simple diffusion, facilitated diffusion, osmosis and filtration are examples of passive transport.

Question no 20: What is the Movement across membrane?

Answer: This takes place when molecules diffuse across the cell membrane by travelling through specific transport proteins. It occurs down a concentration gradient - molecules move from an area of high to lower concentration. This does not require a supply of energy because diffusion is a spontaneous process.

Question no 21: Define Carrier proteins?

Answer: A carrier protein is a type of protein that transports a specific substance through intracellular compartments, into the extracellular fluid, or across cells as opposed to channel proteins, which is another membrane transport protein, that is less selective in transporting molecules. Similar to other membrane transport proteins, carrier proteins are located in lipid bilayers cell structures, such as cell membranes, mitochondria, and chloroplasts.

Carrier proteins bind specific solutes and transfer them across the lipid bilayer by undergoing conformational changes that expose the solute-binding site sequentially on one side of the membrane and then on the other.

Question no 22: Define channel proteins and give their mechanism in the cell membrane?

Answer: A channel protein is a special arrangement of amino acids which embeds in the cell membrane, providing a hydrophilic passageway for water and small, polar ions. Like all transport proteins, each channel protein has a size and shape which excludes all but the most specific molecules.

Most ion channels are gated—that is, they open and close either spontaneously or in response to a specific stimulus, such as the binding of a small molecule to the channel protein (ligand-gated ion channels) or a change in voltage across the membrane that is sensed by charged segments of the channel protein (voltage-gated ion channels).

Question no 23: Define isotropic and anisotopic diffusion?

Answer:

Isotropic diffusion: For most fluids and some homogeneous solid materials like gels, diffusion is the same in every direction. These substances are called isotropic and are characterized by a single diffusion coefficient (D).

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Anisotopic diffusion: Biological tissues, on the other hand, are highly structured and typically have different diffusion coefficients along different directions and are called anisotropic.

Question no 24: Write a note on types of membranes?

Answer:

Permeable membrane: A permeable membrane is a biological or synthetic material that has tiny holes in it allowing small particles to move through it.

Impermeable membrane: Impermeable membranes do not allow any water to pass through them. They are used to retain water within an area or a structure, or to prevent water ingress into a specific area or structure.

Semipermeable membrane: A semi-permeable membrane is a membrane that only allows certain types of particles to move through it under certain conditions.

Question no 25: Define osmosis?

Answer: Osmosis is the spontaneous net movement or diffusion of solvent molecules through a selectively-permeable membrane from a region of high water potential (region of lower solute concentration) to a region of low water potential (region of higher solute concentration), in the direction that tends to equalize the solute concentrations on the two sides.

Question no 26: What are the effects of osmosis in animal cells?

Answer: Red blood cells placed in a solution with a higher water concentration compared to their contents (eg pure water) will gain water by osmosis, swell up and burst.

Water will diffuse from a higher water concentration outside the cell to a lower water concentration inside the cell. Red blood cells placed in a solution with a lower water concentration compared to their contents will lose water by osmosis and shrink.

Water will diffuse from a higher water concentration inside the cell to a lower water concentration outside the cell.

Question no 27: How do bacteria undergo random walk?

Answer: A random walk enables bacteria to search for food and flee from harm. Bacteria propel themselves with the aid of flagella in a process called chemo taxis, and a typical bacteria trajectory has many characteristics of a random walk. They move forward for a certain distance, and then the course is abruptly altered by a process called tumbling. The average change of direction is about 60°.

Question no 28: Define Brownian motion?

Answer: Random motion of gas molecules is called Brownian motion. The Brownian motion of a gas or liquid molecules along a zig – zag is also an example of random motion.

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This pattern of motion typically consists of random fluctuations in a particle's position inside a fluid sub-domain, followed by relocation to another sub-domain. Each relocation is followed by more fluctuations within the new closed volume.

Question no 29: Write down the Principle of Gel Filtration Chromatography?

Answer: Gel-filtration chromatography, also known as size exclusion chromatography, is a versatile technique that permits the separation of proteins and other biological molecules.

Principle

The gel filtration chromatography is based on the molecular size and the hydrodynamic volume of the components. The molecules are separated by the differential exclusion or inclusion of solutes as they pass through the stationary phase containing heterosporous cross-linked polymeric gel or beads. Different permeation rates of the solute molecules cause them to sift in the interior of the gel particles. A column of the porous matrix is in equilibrium with the mobile phase for the separation of the molecules. Large molecules are entirely excluded from the pores and come first in the effluent. Smaller molecules get distributed between the mobile phase and the outside of the sieve. Then, they pass through the column at a slower rate and appear later in the effluent.

Question no 30: Write a note on X-ray Crystallography?

Answer: X-ray crystallography is a tool used for determining the atomic and molecular structure of a crystal. The underlying principle is that the crystalline atoms cause a beam of X-rays to diffract into many specific directions. X-ray crystallography is also a routine technique to determine how a drug interacts with its target and what modifications could improve the interaction. It is a powerful tool in structural biology and can offer insight into structured-based understanding of general anesthetic action on various relevant molecular targets

By measuring the angles and intensities of these diffracted beams, a crystallographer can produce a 3D picture of the density of electrons within the crystal. From this electron density image, the mean positions of the atoms in the crystal can be determined, as well as their chemical bonds, their disorder, and various other information. The method revealed the structure and function of many biological molecules, including vitamins, drugs, proteins, and nucleic acids, such as DNA.

Question no 31: Define Nuclear magnetic resonance (NMR) spectroscopy?

Answer: NMR spectroscopy is an analytical chemistry technique used in quality control and research for determining the content and purity of a sample as well as its molecular structure.

NMR spectroscopy, which is sensitive to the short-range order of all the resonant nuclei in the sample investigated, should be ideal to study nanomaterials, where the long-range order is usually interrupted by the surface and other defect sites [67]. In principle, this spectroscopy technique relies on the interaction between material and electromagnetic radiation. This technique is employed to observe the local magnetic fields around the atomic nuclei of the individual material.

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The use of NMR to determine the three-dimensional structure of a protein has some advantages over X-ray diffraction in that it can be carried out in solution and thus the protein is free of the constraints of the crystal. The two-dimensional NMR techniques generally used are NOESY, which measures the distances between atoms through space, and COESY, which measures distances through bonds.

Question no 32: Define High-performance liquid chromatography (HPLC)?

Answer: High-performance liquid chromatography (HPLC), formerly referred to as high-pressure liquid chromatography, is a technique in analytical chemistry used to separate, identify, and quantify each component in a mixture.

It relies on pumps to pass a pressurized liquid solvent containing the sample mixture through a column filled with a solid adsorbent material. Each component in the sample interacts slightly differently with the adsorbent material, causing different flow rates for the different components and leading to the separation of the components as they flow out of the column.

HPLC has been used for legal, research and medical purposes such as:

- the production process of pharmaceutical and biological products
- detecting performance enhancement drugs in urine
- detecting vitamin D levels in blood serum

Question no 33: Why does hydrogen bonding occur?

Answer: The reason hydrogen bonding occurs is because the electron is not shared evenly between a hydrogen atom and a negatively charged atom. Hydrogen in a bond still only has one electron, while it takes two electrons for a stable electron pair. The result is that the hydrogen atom carries a weak positive charge, so it remains attracted to atoms that still carry a negative charge.

Question no 34: Define Catabolic pathways?

Answer: Catabolic pathways involve the breakdown of complex molecules into simpler ones and typically release energy. Energy stored in the bonds of complex molecules, such as glucose and fats, is released in catabolic pathways. It's then harvested in forms that can power the work of the cell (for instance, through the synthesis of ATP).

Question no 35: What is Open system?

Answer: An open system is a system that has external interactions. Such interactions can take the form of information, energy, or material transfers into or out of the system boundary.

Question no 36: Name any two polar compounds?

Answer:

- Water (H2O)

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- Hydrogen Fluoride (HF)
- Ethanol (CH3CH2OH)

Question no 37: Name two metabolic pathways?

Answer: Metabolic pathways can be broadly divided into two categories based on their effects. Photosynthesis, which builds sugars out of smaller molecules, is a "building up," or anabolic, pathway. In contrast, cellular respiration breaks sugar down into smaller molecules and is a "breaking down," or catabolic, pathway.

Question no 38: Difference homo-nuclear and hetero-nuclear diatomic molecules?

Answer: The key difference between homonuclear and heteronuclear diatomic molecules is that homonuclear diatomic molecules contain two atoms of the same element whereas heteronuclear diatomic molecules contain two atoms of different elements.

Homonuclear diatomic molecules are substances composed of two atoms of the same chemical element bonded to each other via a covalent bond(s).

Heteronuclear diatomic molecules are substances composed of two atoms of two different chemical elements bonded to each other via a covalent bond(s). Therefore, the atoms of the heteronuclear diatomic molecule are different from each other.

Question no 39: What is Cardiac Electrophysiology?

Answer: Cardiac electrophysiology is a branch of cardiology and basic science focusing on the electrical activities of the heart. The term is usually used in clinical context, to describe studies of such phenomena by invasive (intracardiac) catheter recording of spontaneous activity as well as of cardiac responses to programmed electrical stimulation - clinical cardiac electrophysiology.

Question no 40: Write the biophysics of membrane and give importance of biology?

Answer: Membrane biophysics is the study of the physical principles governing biological membranes, including lipid-raft formation and protein–lipid coupling, as well as their mechanical characteristics, and the effect they have on Para cellular transport and phenomena relating to cell shape.

- Membrane biophysics is the study of biological membrane structure and function using physical, computational, mathematical, and biophysical methods.
- Bridge between biology and physics.
- Study of biological systems and biological processes using physics-based methods or based on physical principles.
- Biophysics discovers how atoms are arranged to work in DNA and proteins.

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- Biophysicists are discovering how proteins work; they use these protein structures for discovering how biological machines work, in health and also in diseases.

Question no 43: Define van der Waals force?

Answer: The relatively weak attractive forces that act on neutral atoms and molecules and that arise because of the electric polarization induced in each of the particles by the presence of other particles.

Question no 44: Describe enthalpy and entropy?

Answer: **Enthalpy:**

The sum of the internal energy and the product of the pressure and volume of a thermodynamic system is called enthalpy. It is an energy-like property or state function—it has the dimensions of energy (and is thus measured in units of joules or ergs), and its value is determined entirely by the temperature, pressure, and composition of the system and not by its history.

In symbols, the enthalpy, H, equals the sum of the internal energy, E, and the product of the pressure, P, and volume, V, of the system: $H = E + PV$.

Entropy:

The measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of entropy is also a measure of the molecular disorder, or randomness, of a system.

The units of entropy are J/K. The temperature in this equation must be measured on the absolute, or Kelvin temperature scale.

Question no 45: How protein induces membrane curvature?

Answer: Sufficiently rigid scaffold proteins can bend membranes using a membrane-adherent surface that causes the membrane to follow the contour of the protein. Proteins that insert into the lipid bilayer can bend membranes by changing the spontaneous curvature of the local monolayer.

Question no 46: Describe manometer, barometer?

Answer: A Manometer is a device to measure pressures. A common simple manometer consists of a U shaped tube of glass filled with some liquid. Typically, the liquid is mercury because of its high density.

A barometer is a scientific instrument used to measure atmospheric pressure, also called barometric pressure.

Question no 47: Application of biophysics in biology?

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Answer: Biophysics has been essential to the development of many life-saving treatments and devices including kidney dialysis, radiation therapy, cardiac defibrillators, pacemakers, and artificial heart valves.

Question no 48: What is thermodynamics and what are four laws of thermodynamics?

Answer: There are four laws of thermodynamics. They talk about temperature, heat, work, and entropy. They are used in thermodynamics and other sciences, for example chemistry.

Thermodynamics has three main laws: the first law, the second law, and the third law. Then there was another law, called the "zeroth law." The law of conservation of mass is also an important idea in thermodynamics, but it is not called a law.

Question no 49: What is Membrane fluidity?

Answer: In biology, membrane fluidity refers to the viscosity of the lipid bilayer of a cell membrane or a synthetic lipid membrane. Lipid packing can influence the fluidity of the membrane. Viscosity of the membrane can affect the rotation and diffusion of proteins and other bio-molecules within the membrane, there-by affecting the functions of these things.

Membrane fluidity is affected by fatty acids. More specifically, whether the fatty acids are saturated or unsaturated has an effect on membrane fluidity.

Question no 50: What is Gibbs free energy?

Answer: The energy associated with a chemical reaction that can be used to do work

- The Relation between Entropy ΔS and Enthalpy ΔH.
- The equation is □G = □H - T□S

Question no 51: Describe Secondary Structure of protein?

Answer: Secondary structure is formally defined by the pattern of hydrogen bonds between the amino hydrogen and carboxyl oxygen atoms in the peptide backbone. Secondary structure may alternatively be defined based on the regular pattern of backbone dihedral angles in a particular region of the Ramachandran plot regardless of whether it has the correct hydrogen bonds.

Question no 52: Discuss alpha-helix and beta sheet?

Answer: **Alpha-helix:** The backbone follows a helical path (right, left), flexible Hydrogen bonds between backbone amino and carbonyl groups and those in the next turn of the helix. The R-groups protrude out from the helix Proline tends to interrupt an alpha-helix

Beta sheet: Strands of protein lie adjacent to one another, interacting laterally via hydrogen bonds, between carbonyl oxygen and amino H atoms Successive side chains point straight up, then straight down.

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Question no 53: Define exergonic and endergonic reaction?

Answer: The terms endergonic and exergonic are used to explain two types of chemical reactions.

An **endergonic reaction** is a non-spontaneous reaction. It does not occur in normal conditions such as at room temperature and atmospheric pressure. An exergonic reaction is the opposite of endergonic reaction.

An **exergonic reaction** is a spontaneous reaction. It occurs at normal conditions without any external force. Each and every chemical reaction can be categorized as an endergonic reaction or an exergonic reaction. The main difference between endergonic and exergonic is that endergonic reaction requires energy from the outside whereas exergonic reactions release energy to the outside.

Question no 54: Factors that increase entropy?

Answer: The entropy of a substance increases with its molecular weight and complexity and with temperature. The entropy also increases as the pressure or concentration becomes smaller. Entropies of gases are much larger than those of condensed phases.

Question no 55: What is the function of cell membrane?

Answer: Cell membranes serve as barriers and gatekeepers. They are semi-permeable, which means that some molecules can diffuse across the lipid bilayer but others cannot. Small hydrophobic molecules and gases like oxygen and carbon dioxide cross membranes rapidly. Small polar molecules, such as water and ethanol, can also pass through membranes, but they do so more slowly.

Question no 56: Order and disorder in biological system?

Answer:

- Entropy is a measure of disorder: cells are NOT disordered and so have low entropy.
- The flow of energy maintains order and life. Entropy wins when organisms cease to take in energy and die.
- The Second Law of Thermodynamics states that the disorder, or entropy, in the universe is increasing. This also means that conversions of energy from one form to another are inefficient, with some energy being lost to cells as HEAT.

Question no 57: What are factors that change entropy? (3)

Answer: **Factors can increase Entropy**

1) Adding more particles increases the collisions and the randomness of motion

2) Increasing temperature: The temperature can increase the velocity of particle motions

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3) As volume increases the particles in a system allowed to roam in greater space; thus, more random motion

4) Increasing the freedom of motions of atoms and molecules - In general, the entropy of a system increases for processes in which: - Gases form from either solids or liquids - Liquids or solutions form from solids

Question no 58: What is spontaneity? Describe the factors that contribute to spontaneity?

Answer: The tendency for a process to advance to equilibrium without external influence something that happens naturally is spontaneous Some physical and chemical processes proceed naturally with no apparent cause or stimulus Any process will be spontaneous in one direction The reverse is non-spontaneous

• Change in enthalpy (ΔH) • Change in entropy (ΔS)

• Temperature of the reaction (T)

Question no 59: What are macroscopic and microscopic approaches of thermodynamics?

Answer: Behavior of matter can be studied by these two levels:

1. Macroscopic approach— (Macro mean big or total)

2. Microscopic approach— (Micro means small)

Macroscopic approach: Certain quantity of matter is considered, without a concern on the events occurring at the molecular level. These effects can be perceived by human senses or measured by instruments. e.g: pressure, temperature Approach to thermodynamics is concerned with gross or overall behavior. This is known as classical thermodynamics.

Microscopic approach: The effect of molecular motion is Considered. e.g: at microscopic level the pressure of a gas is not constant, the temperature of a gas is a function of the velocity of molecules. A more elaborate approach, based on the average behavior of large groups of individual particles, is called statistical thermodynamics dipole-dipole forces of attraction (Hydrogen Bonding's) Two polar molecules Positively charged portion of one molecule is attracted to the negatively charged portion of another molecule One molecule polar and other molecule not to be polar.

Question no 60: Define membrane fluidity?

Answer: Biological membranes are fluid dynamic that separate the inside from the outside of cells One remarkable feature of all biological membranes is their flexibility-their ability to change shape without losing their integrity and becoming leaky Lipid bilayer structure is quite stable.

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Cell membrane: the thin and flexible boundary of a cell (also called a plasma membrane) It separates a cell from its surroundings. It allows nutrients to enter and waste to be removed.

The fluidity of the lipid bilayer is vital for the function of the membrane Membranes composed of phospholipids are highly mobile

Membrane fluidity is influenced by:

- Temperature
- Composition

Question no 61: Define thermodynamics property? Explain intensive and extensive property? (5)

Answer:

A measurable quantity that defines the condition of a system e.g. Temperature T, Mass m, P Pressure, Volume V, Density. Another useful property is specific volume defined as the volume of a unit mass

Types of Properties Thermodynamic properties are divided into two broad types:

❖ Intensive properties

❖ Extensive properties

Extensive properties: Extensive properties are those whose values depend on the size-or extent-of the system.

- Mass m,
- Volume V,
- Total energy E These are examples of extensive properties.

If mass is increased, the value of extensive property also increases

Intensive properties

Intensive properties are those that are independent of the size of a system, such as

- temperature,
- pressure,
- Density

Independent of mass

➢ Extensive properties per unit mass are called specific properties. Some examples of specific properties are o specific volume (v = V/m), o specific total energy (e = Elm), and o specific internal energy (u = V/m). An example is specific volume, which has units of volume/mass, typically expressed as cubic feet per pound or cubic meters per kilogram

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Question no 62: Application of biophysics at biological level? helix, beta-sheets in secondary structure of protein?

Answer: **Secondary structure: Alpha-helix** The backbone follows a helical path (right, left), flexible Hydrogen bonds between backbone amino and carbonyl groups and those in the next turn of the helix. The R-groups protrude out from the helix Proline tends to interrupt an alpha-helix

Secondary structure: Beta sheet Strands of protein lie adjacent to one another, interacting laterally via hydrogen bonds, between carbonyl oxygen and amino H atoms Successive side chains point straight up, then straight down.

Question no 63: What is viscosity? 2

Answer: Viscosity is the measure of resistance of the fluid to flow and it differs from one fluid to other.

Question no 64: Write 3 types of spectroscopy? 3

Answer:

- Atomic Emission spectroscopy
- Atomic Absorption spectroscopy
- Scattering spectroscopy

Question no 65: ADP+Pi complete the reaction? 3

Answer: ADP + P i →ATP + H 2 O

G° = 29 kJ/mol (Endergonic reaction)

Question no 66: What is covalent bond? Explain. 5

Answer: A covalent bond, also called a molecular bond, is a chemical bond that involves the sharing of electron pairs between atoms.

A covalent bond consists of the mutual sharing of one or more pairs of electrons between two atoms. These electrons are simultaneously attracted by the two atomic nuclei. A covalent bond forms when the difference between the electronegativity's of two atoms is too small for an electron transfer to occur to form ions. Shared electrons located in the space between the two nuclei are called bonding electrons. The bonded pair is the "glue" that holds the atoms together in molecular units.

Question no 67: How substance moves in and out the cell? 5

Answer:

- Osmosis is a type of diffusion that involves the random movement of water molecules.

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- It is the movement of water across a partially-permeable membrane, from an area of high water concentration to an area of low water concentration.
- Osmosis explains how water moves in and out of animal cells through the cell membrane.
- If water continues to leave a cell it will shrink and become crinkly. If water continues to enter a cell it will eventually burst.
- These things happen because animal cells are not surrounded by an inelastic cell wall, unlike plant cells.
- Animal cells do not have cell walls.
- In hypotonic solutions, animal cells swell up and explode as they cannot become turgid because there is no cell wall to prevent the cell from bursting.
- When the cell is in danger of bursting, organelles called contractile vacuoles will pump water out of the cell to prevent this.
- In hypertonic solutions, water diffuses out of the cell due to osmosis and the cell shrinks. Thus, the animal cell has always to be surrounded by an isotonic solution.
- In the human body, the kidneys provide the necessary regulatory mechanism for the blood plasma and the concentration of water and salt removed from the blood by the kidneys is controlled by a part of the brain called the hypothalamus.
- The process of regulating the concentration of water and mineral salts in the blood is called osmoregulation.
- Animals which live on dry land must conserve water and so must animals which live in the salty sea water, but animals which live in freshwater have the opposite problem; they must get rid of excess water as fast as it gets into their bodies by osmosis.

Question no 68: Write a note on sliding friction. Write any example? (5)

Answer: Sliding friction occurs between objects as the slide against each other. It acts in the direction opposite to the direction of motion. It prevents the object from moving too fast. It is also called as kinetic friction.

• Examples: A book sliding from an inclined desk, A kid sliding on a slide.

• When sliding friction is acting there must be another force existing to keep the body in moving, in case of the book sliding from the desk the other force acting is gravitational force.

Question no 69: Mention three part of mass spectrometer? 3

Answer: Mass spectrometers can be divided into three fundamental parts, namely

- Ionization source
- Analyzer
- Detector

Question no 70: Why does a hydrogen bonding occur? 3

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Answer: Since the hydrogen donor is strongly electronegative, it pulls the covalently bonded electron pair closer to its nucleus, and away from the hydrogen atom. The hydrogen atom is then left with a partial positive charge, creating a dipole-dipole attraction between the hydrogen atom bonded to the donor, and the lone electron pair on the acceptor. This results in a hydrogen bond.

Question no 71: Name two types of hydrogen bond? 2

Answer:

1. Intramolecular hydrogen bonds
2. Intermolecular hydrogen bonds

Question no 72: Define rotatory motion? 2

Answer: Rotatory motion is the motion that occurs when a body rotates on its own axis. A few examples of the rotatory motion are as follows: The motion of the earth about its own axis around the sun is an example of rotary motion.

Question no 73: Explain the hydrodynamics?

Answer: The scientific study of the motion of fluids, especially non-compressible liquids, under the influence of internal and external forces.

- The behavior of large biomolecules—proteins, carbohydrates, and nucleic acid in solution is complex and directly related to molecular size, shape, and flexibility.
- Analysis of hydrodynamic behavior thus provides important information about the structure, dynamics, and interactions of bio macromolecules.

Question no 74: Random motion of moving faster molecule RNA function? or characteristics of channel protein?

Answer: Integral lipoproteins which contain a pore via which ions may cross from one side of the membrane to the other

- Channel proteins are ion-selective and may be gated to regulate the passage of ions in response to certain stimuli
- Channel proteins only move molecules along a concentration gradient (i.e. are not used in active transport)
- Channel proteins have a much faster rate of transport than carrier proteins.

Question no 75: State equation of Enthalpy? 2 marks

Answer: The enthalpy of a system is equal to the system's internal energy plus the product of its pressure and volume.

$H = E + PV$

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Question no 76: Name two main methods of Frictationations? 2 marks

Answer:

- Bioassay-guided fractionation
- Blood fractionation

Question no 77: Write above spin coupling? 5 marks

Answer: The effective magnetic field is also affected by the orientation of neighboring nuclei. This effect is known as spin-spin coupling which can cause splitting of the signal for each type of nucleus into two or more lines. The size of the splitting (coupling constant or J) is independent of the magnetic field and is therefore measured as an absolute frequency (usually Hertz). The number of splitting indicates the number of chemically bonded nuclei in the vicinity of the observed nucleus.

Question no 78: Define photosynthesis (2)

Answer: In photosynthesis, the energy is supplied by the sun. Light energy is absorbed by cells in plant leaves and converted to chemical energy.

Question no 79: Who is Karl Pearson?

Answer: An English mathematician and biostatistician. In 1892 he published The Grammar of Science, in which he argued that the scientific method is essentially descriptive rather than explanatory. Soon he was making the same argument about statistics, emphasizing especially the importance of quantification for biology, medicine, and social science.

Question no 80: What is Ionic bond ?2

Answer: Ionic bonding is a type of chemical bond that involves the electrostatic attraction between oppositely charged ions.

Question no 81: Define these terms: Hypotonic, Hypertonic, isotonic?3

Answer:

- If the water concentration of the cells cytoplasm is lower than that of the medium (i.e. the medium is a hypotonic solution
- If the water concentration inside the cell is higher than that of the medium is a hypertonic solution
- If the water concentration inside the cell is the same as that in the surrounding medium is a isotonic solution

Question no 81: NMR instruments details? (5)

Answer:

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1. **Sample holder:** - Glass tube with 8.5 cm long,0.3 cm width
2. **Permanent magnet:** - It provides homogeneous magnetic field at 60-100 MHZ
3. **Magnetic coils:** -These coils induce magnetic field when current flows through them.
4. **Sweep generator:** - To produce the equal amount of magnetic field pass through the sample
5. **Radio frequency (radio transmitter coil):** - A radio transmitter coil that produces a short powerful pulse of radio waves
6. **Radio frequency (radio receiver coil):** - A radio receiver coil that detects Receiver radio frequencies emitted as nuclei relax to a lower energy level
7. **Readout system:** - A computer that analyses and record the data

Question no 82: Write Enthalpy units?

Answer: The unit of measurement for enthalpy in the International System of Units (SI) is the **joule.** Other historical conventional units still in use include the British thermal unit (BTU) and the **calorie.**

Question no 83: Name of main energy carriers by cell?

Answer: The main chemical energy carrier of the cell is ATP, short for adenosine triphosphate, and cells convert energy from ATP to many things.

Question no 84: What is millisievert?

Answer: The millisievert and milligray as measures of radiation dose and exposure. In the SI system, a millisievert (mSv) is defined as "the average accumulated background radiation dose to an individual for 1 year, exclusive of radon, in the United States.

Question no 85: Name the voltage gated Channels

Answer: Voltage-gated ion-channels are usually ion-specific, and channels specific to sodium (Na+), potassium (K+), calcium (Ca2+), and chloride (Cl–) ions have been identified. And proton ion channel also.

Question no 86: What three spectroscopy techniques

Answer: **Atomic Emission spectroscopy**

It uses the range of electromagnetic spectra in which a substance radiates (emits). The substance first must absorb energy. This energy can be from a variety of sources, which determines the name of the subsequent emission, like luminescence. Molecular luminescence techniques include spectro-flourimetry

Atomic Absorption spectroscopy

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It uses the range of the electromagnetic spectra in which a substance absorbs. This includes atomic absorption spectroscopy and various molecular techniques, such as infra-red spectroscopy in that region and Nuclear Magnetic resonance spectroscopy in the radio region

Scattering spectroscopy

It uses the range of electromagnetic spectra in which a substance radiates (emits). The substance first must absorb energy. This energy can be from a variety of sources, which determines the name of the subsequent emission, like luminescence. Molecular luminescence techniques include spectro-flourimetry.

Question no 87: What is Biophysical society?

Answer: 1957: Biophysical Society was founded

Aims: to boost the development and distribution of knowledge in biophysics through many activities including meetings, publications, community outreach, and career placement Members: >9,000, work in academia, industry, and government agencies worldwide Annual Meeting is held each year in February or March. With over 7,000 attendees The Society publishes the Biophysical journal twice monthly

Question no 88: What is the direct effect of ionizing radiations?

Answer: The direct effect of ionizing radiation and the radiant energy is immediately absorbed inside the cellular nucleus.

Question no 89: Explain types of HPLC

Answer: **Normal Phase HPLC:** This method separates analytes on the basis of polarity. NP-HPLC uses polar stationary phase and non-polar mobile phase. Therefore, the stationary phase is usually silica and typical mobile phases are hexane, methylene chloride, chloroform, diethyl ether, and mixtures of these. Polar samples are thus retained on the polar surface of the column packing longer than less polar materials.

Reverse Phase HPLC: The stationary phase is nonpolar (hydrophobic) in nature, while the mobile phase is a polar liquid, such as mixtures of water and methanol or acetonitrile. It works on the principle of hydrophobic interactions hence the more nonpolar the material is, the longer it will be retained.

Size-exclusion HPLC: The column is filled with material having precisely controlled pore sizes, and the particles are separated according to its their molecular size. Larger molecules are rapidly washed through the column; smaller molecules penetrate inside the porous of the packing particles and elute later.

Ion-Exchange HPLC: The stationary phase has an ionically charged surface of opposite charge to the sample ions. This technique is used almost exclusively with ionic or ionizable samples. The stronger the charge on the sample, the stronger it will be attracted to the ionic surface and

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thus, the longer it will take to elute. • The mobile phase is an aqueous buffer, where both pH and ionic strength are used to control elution time.

Question no 90: Explain third law of thermodynamics (10)

Answer: The third law of thermodynamics was developed by the German chemist Walther Nernst during the years 1906–12.

The third law of thermodynamics is often referred to as Nernst's theorem or Nernst's postulate.

Nernst Heat Theorem (a consequence of the Third Law) is: It is impossible for any process, no matter how idealized, to reduce the entropy of a system to its absolute-zero value in a finite number of operations.

An alternative version of the third law of thermodynamics as stated by Gilbert N. Lewis and Merle Randall in 1923:

"If the entropy of each element in some (perfect) crystalline state be taken as zero at the absolute zero of temperature, every substance has a finite positive entropy; but at the absolute zero of temperature the entropy may become zero, and does so become in the case of perfect crystalline substances."

The third law of thermodynamics states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.

The entropy of a system at absolute zero is typically zero, and in all cases is determined only by the number of different ground states it has. Specifically, the entropy of a pure crystalline substance (perfect order) at absolute zero temperature is zero. This statement holds true if the perfect crystal has only one state with minimum energy. It says that when we are considering a totally perfect (100% pure) crystalline structure, at absolute zero (0 Kelvin), it will have no entropy (S).

Note that if the structure in question were not totally crystalline, then although it would only have an extremely small disorder (entropy) in space, we could not precisely say it had no entropy. The entropy of a pure perfect crystal is zero (0) at zero Kelvin (0° K). Simply putting, the third law of thermodynamics states that if all the thermal motion of molecules (kinetic energy) could be removed, a state called absolute zero would occur. The Universe will attain absolute zero when all energy and matter is randomly distributed across space. The current temperature of empty space in the Universe is about 2.7 Kelvins

Question no 91: Shortly explain electro dialysis (3)

Answer: Electro dialysis (ED) is used to transport salt ions from one solution through ion exchange membranes to another solution under the influence of an applied electric potential difference.

Question no 92: Name two polar compounds?

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Answer:

- H-Cl
- H-I

Question no 93: Define motion? 2

Answer: In physics, motion is a change in position of an object over time. • Motion is described in terms of displacement, distance, velocity, acceleration, time, and speed. • Motion of a body is observed by attaching a frame of reference to an observer and measuring the change in position of the body relative to that frame.

Question no 94: Explain third law of motion?

Answer: Whenever one body exerts a force F onto a second body, (in some cases, which is standing still) the second body exerts the force −F on the first body. F and −F are equal in magnitude and opposite in sense. So, the body which exerts F will go backwards

Question no 95: Explain intermolecular Hydrogen bonding?5

Answer: A hydrogen bond is an intermolecular force (IMF) that forms a special type of dipole-dipole attraction when a hydrogen atom bonded to a strongly electronegative atom exists in the vicinity of another electronegative atom with a lone pair of electrons. Intermolecular forces (IMFs) occur between molecules. Other examples include ordinary dipole-dipole interactions and dispersion forces. Hydrogen bonds are are generally stronger than ordinary dipole-dipole and dispersion forces, but weaker than true covalent and ionic bonds.

Current papers 2022

1. Name of 4 non polar Amino acid?

Answer: The non-polar amino acids (shown here) include: alanine, cysteine, glycine, isoleucine, leucine, methionine, phenylalanine, proline, tryptophan, tyrosine and valine.

2. Structure of amino acids?

Answer: An amino acid is an organic molecule that is made up of a basic amino group (−NH2), an acidic carboxyl group (−COOH), and an organic R group (or side chain) that is unique to each amino acid. The term amino acid is short for α-amino [alpha-amino] carboxylic acid.

3. Commonly used Applications of NMR?

Answer: Nuclear magnetic resonance spectroscopy (NMR) is the most powerful technique for determining the structure of organic compounds. NMR techniques are used successfully

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in various food systems for quality control and research. NMR spectroscopy is used to determine structure of proteins, amino acid profile, carotenoids, organic acids, lipid fractions, the mobility of the water in foods. NMR spectroscopy is also used to identify and quantify the metabolites in foods. Also vegetable oils, fish oils, fish and meat, milk, cheese, wheat, fruit juices, coffee, green tea, foods such as wine and beer are among the last NMR applications.

4. Write a short note of NMR and imaging?

Answer: A procedure that uses radio waves, a powerful magnet, and a computer to make a series of detailed pictures of areas inside the body. A contrast agent, such as gadolinium, may be injected into a vein to help the tissues and organs show up more clearly in the picture. Nuclear magnetic resonance imaging may be used to help diagnose disease, plan treatment, or find out how well treatment is working.

It is especially useful for imaging the brain and spinal cord, the heart and blood vessels, the bones, joints, and other soft tissues, the organs in the pelvis and abdomen, and the breast. Also called magnetic resonance imaging, MRI, and NMRI.

5. What is function of protein?

Answer: Proteins are made up of many building blocks, known as amino acids. Our body needs dietary protein to supply amino acids for the growth and maintenance of our cells and tissues. Our dietary protein requirement changes throughout life. Five major functions of proteins in human body are following:

- Repair and Maintenance
- Hormone Production.
- Enzymes.
- Immune Function.
- Energy

6. Define electrophoresis?

Answer: Electrophoresis is a laboratory technique used to separate DNA, RNA or protein molecules based on their size and electrical charge. An electric current is used to move the molecules through a gel or other matrix. Pores in the gel or matrix work like a sieve, allowing smaller molecules to move faster than larger molecules. To determine the size of the molecules in a sample, standards of known sizes are separated on the same gel and then compared to the sample.

7. Write the applications of NMR?

Answer: Nuclear magnetic resonance spectroscopy (NMR) is the most powerful technique for determining the structure of organic compounds. NMR techniques are used successfully in various food systems for quality control and research.

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NMR spectroscopy is used to determine structure of proteins, aminoacid profile, carotenoids, organic acids, lipid fractions, the mobility of the water in foods. NMR spectroscopy is also used to identify and quantify the metabolites in foods.

8. What is uniqueness in Biophysics?

Answer:

1. Biophysics is a growing enterprise world-wide, driven primarily by the widespread realization of the major contributions made to biological science by a combination of truly state-of-the-art physical measurements with modern molecular biology.

2. The field occupies a unique and central position at the intersection of the biological, chemical, physical, and computational sciences

3. Development and advancement of new techniques and approaches.

4. Biophysicists as a group most often develop the novel, sophisticated experimental methods that reveal molecular level details with unique clarity.

9. Spontaneous and non-spontaneous reactions?

Answer:

Spontaneous Process:

Spontaneity does not entail that the reaction happens quickly. The pace of a reaction is determined by the chemical kinetics of the reaction rather than its spontaneity. In a spontaneous process, every reactant tends to generate the corresponding product. This pattern is linked to stability.

Non-Spontaneous Process:

A chemical reaction known as an endergonic reaction (also known as a nonspontaneous reaction) is one in which the standard change in free energy is positive, and energy is absorbed. Because the whole quantity of energy represents a loss, the net outcome is negative.

Endergonic processes can also be driven by connecting them through a common intermediate to a highly exergonic reaction.

10.Force of static friction?

Answer: In static friction, the frictional force resists force that is applied to an object, and the object remains at rest until the force of static friction is overcome. In kinetic friction, the frictional force resists the motion of an object.

11.Diffusion example in plants?

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Answer: The examples of diffusion in plants are:

- The uptake of water and nutrients from the soil by the roots
- The distribution of nutrients and water to all the parts of the plant
- Diffusion of Carbon Dioxide into the leaf of the plants during photosynthesis
- Release of oxygen out of the stomata of the leaf

12.Discovery of DNA?

Answer: The discovery in 1953 of the double helix, the twisted-ladder structure of deoxyribonucleic acid (DNA), by James Watson and Francis Crick marked a milestone in the history of science and gave rise to modern molecular biology, which is largely concerned with understanding how genes control the chemical processes within cells.

13. Factors considered during gel filtration?

Answer: Various factors should be considered when designing a gel-filtration system. These include:

- matrix choice,
- sample size and concentration,
- column parameters,
- choice of eluent,
- effect of flow rate
- Column cleaning and storage.

14.Entropy? (2)

Answer: Entropy, the measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of entropy is also a measure of the molecular disorder, or randomness, of a system. The concept of entropy provides deep insight into the direction of spontaneous change for many everyday phenomena.

15.Osmotic pressure? (2)

Answer: Osmotic pressure can be defined as the minimum pressure that must be applied to a solution to halt the flow of solvent molecules through a semipermeable membrane (osmosis). It is a colligative property and is dependent on the concentration of solute particles in the solution. Osmotic pressure can be calculated with the help of the following formula:

$\pi = iCRT$

Where,

π is the osmotic pressure

i is the van't Hoff factor

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C is the molar concentration of the solute in the solution

R is the universal gas constant

T is the temperature

16.Thermal and mechanical equilibrium (3)

Answer: In thermodynamics, equilibrium refers to a state of equilibrium with respect to all possible changes, thermal, mechanical and chemical.

Thermal equilibrium

A state of thermal equilibrium can be described as one in which the temperature of the system is uniform.

Mechanical equilibrium

Mechanical equilibrium means there is no unbalanced force. In other words, there is no pressure gradient within the system.

17.What is biology?

Answer: Biology is a branch of science that deals with living organisms and their vital processes. Biology encompasses diverse fields, including botany, conservation, ecology, evolution, genetics, marine biology, medicine, microbiology, molecular biology, physiology, and zoology.

18.State 3 laws of dry friction

Answer: The elementary property of sliding (kinetic) friction was discovered by experiment in the 15th to 18th centuries and was expressed as three empirical laws:

Amontons' First Law: The force of friction is directly proportional to the applied load.

Amontons' Second Law: The force of friction is independent of the apparent area of contact.

Coulomb's Law of Friction: Kinetic friction is independent of the sliding velocity.

19.State five functions of cell membrane?

Answer: The following points highlight the top five functions of plasma membrane. The functions are:

- Providing a Selectively Permeable Bar-rier
- Transporting Solutes
- Transporting Macromolecules
- Responding to External Signals

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- Intercellular Interaction

20.What is electro dialysis?

Answer: Electro dialysis (ED) is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference. This is done in a configuration called an electro dialysis cell.

21.Give an example of ion dipole force?

Answer: The example of ion- dipole interaction is the solution of table salt (NaCl) in water. Sodium chloride undergoes dissociation to form Na+ and Cl– ions and water is polar solvent. The positive sodium ion will be attracted to the slightly negative oxygen atoms in the water molecule, while the negative chloride ion is attracted to the slightly positive hydrogen atoms.

22.Surface tension?

Answer: Surface tension is the tendency of liquid surfaces at rest to shrink into the minimum surface area possible. Surface tension is what allows objects with a higher density than water such as razor blades and insects (e.g. water striders) to float on a water surface without becoming even partly submerged.

23.What are the different types of cell membrane? (2)

Answer: **Permeable membrane:** A permeable membrane is a biological or synthetic material that has tiny holes in it allowing small particles to move through it.

Impermeable membrane: Impermeable membranes do not allow any water to pass through them. They are used to retain water within an area or a structure, or to prevent water ingress into a specific area or structure.

Semipermeable membrane: A semi-permeable membrane is a membrane that only allows certain types of particles to move through it under certain conditions.

24. Why cell membrane is called as selectively permeable membrane (3)

Answer:

- The cell membrane is called selectively permeable as it only allows specific molecules to pass.
- Only specific molecules like water and gaseous molecules can pass through the cell membrane directly.
- It stops the flow of other molecules towards the two sides.

25.Define anion and cation?

Answer: A cation is an ion with positive charge, which means it has more protons (positively-charged particles) than electrons (negatively-charged particles). Cations are

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formed when an atom loses one or more electrons: the loss of the negatively-charged electron(s) results in an overall positive charge.

An anion is an ion with negative charge, meaning it has more electrons than protons. Anions are formed when an atom gains one or more electrons: the gain of the negatively-charged electron(s) results in an overall negative charge.

26.What is random walk?

Answer: Random walk, in probability theory, a process for determining the probable location of a point subject to random motions, given the probabilities (the same at each step) of moving some distance in some direction.

27.What is preparative centrifuge?

Answer: Preparative centrifugation refers to a high-velocity centrifuge used in the separation of small submicroscopic particles. It separates small particles such as viruses, viral particles, protein complexes, proteins, lipoproteins, RNA, and plasmid DNA.

The preparative centrifugation can also be used as a density gradient centrifugation method. The sucrose gradient can be used to separate cellular organelles. Caesium gradient can be used to separate nucleic acids.

28.How diffusion occurs in plant leafs?

Answer: Plants use up carbon dioxide during photosynthesis. The concentration of carbon dioxide inside a leaf is therefore low during photosynthesis. The higher concentration of carbon dioxide in the air around the plant causes the gas to move into the leaf by diffusion. It diffuses through small pores called stomata. The stomata are mainly in the lower epidermis (the underneath surface) of leaves.

During photosynthesis, oxygen levels inside the leaf increase. This causes oxygen to diffuse out of the leaf into the air.

29.London dispersion forces in halogen family

Answer: Although charges are usually distributed evenly between atoms in non-polar molecules, spontaneous dipoles can still occur. When this occurs, non-polar molecules form weak attractions with other non-polar molecules. These London dispersion forces are often found in the halogens (e.g., F2 and I2), the noble gases (e.g., Ne and Ar), and in other non-polar molecules, such as carbon dioxide and methane. London dispersion forces are part of the van der Waals forces, or weak intermolecular attractions.

30.Extensive properties (2)

Answer: The extensive properties are defined as the properties which depends on the amount of matter present. Examples are volume, mass, internal energy etc. From the options give

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below, temperature, viscosity & molar heat capacity are intensive properties while internal energy is an extensive property.

31.Two examples of Brownian motion (2)

Answer: Most examples of Brownian motion are transport processes that are affected by larger currents, yet also exhibit pedesis.

Examples include:

- The motion of pollen grains on still water
- Movement of dust motes in a room (although largely affected by air currents)
- Diffusion of pollutants in the air

32.Anabolic Pathway (3)

Answer: Anabolic pathways build complex molecules from simpler ones and typically need an input of energy. Building glucose from carbon dioxide is one example. Other examples include the synthesis of proteins from amino acids, or of DNA strands from nucleic acid building blocks (nucleotides). These biosynthetic processes are critical to the life of the cell, take place constantly, and use energy carried by ATP and other short-term energy storage molecules.

33.Name the Components of Nucleotide (3)

Answer: A nucleotide is the basic building block of nucleic acids (RNA and DNA). A nucleotide consists of a sugar molecule (either ribose in RNA or deoxyribose in DNA) attached to a phosphate group and a nitrogen-containing base. The bases used in DNA are adenine (A), cytosine (C), guanine (G) and thymine (T). In RNA, the base uracil (U) takes the place of thymine. DNA and RNA molecules are polymers made up of long chains of nucleotides.

34.Write the function of RNA?

Answer: The primary function of RNA is to create proteins via translation. RNA carries genetic information that is translated by ribosomes into various proteins necessary for cellular processes. mRNA, rRNA, and tRNA are the three main types of RNA involved in protein synthesis. RNA also serves as the primary genetic material for viruses. Other functions include RNA editing, gene regulation, and RNA interference. These processes are carried out by a group of small regulatory RNAs, which include small nuclear RNA, microRNA, and small interfering RNA.

35.Explain oscillatory motion?

Answer: Oscillatory motion is defined as the to and fro motion of an object from its mean position. The ideal condition is that the object can be in oscillatory motion forever in the

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absence of friction but in the real world, this is not possible and the object has to settle into equilibrium.

Some examples of oscillatory motion are:

- Oscillation of simple pendulum.
- Vibrating strings of musical instruments is a mechanical example of oscillatory motion.
- Movement of spring.

36.Why sliding friction is less than static friction?

Answer: Explanation:

- For a given two surfaces, sliding friction is slightly lesser than that of static friction
- Because it is easier to move the object that is already in motion than the object at rest.
- In other words, if we want to push a car at rest, we need more force than to push a car that is already in motion.

37.What is covalent bond?

Answer: A covalent bond consists of the mutual sharing of one or more pairs of electrons between two atoms. These electrons are simultaneously attracted by the two atomic nuclei. A covalent bond forms when the difference between the electronegativities of two atoms is too small for an electron transfer to occur to form ions.

38.Write the components of cell membrane?

Answer: The cell membrane or plasma membrane is made up of phospholipid bi layer. The main constituents of a cell membrane are lipids, proteins and carbohydrates attached to proteins and lipids

39.Write a note on Enzymes?

Answer: An enzyme is a biological catalyst and is almost always a protein. It speeds up the rate of a specific chemical reaction in the cell. The enzyme is not destroyed during the reaction and is used over and over. A cell contains thousands of different types of enzyme molecules, each specific to a particular chemical reaction.

40.What is Zeroth law of thermodynamics?

Answer: The zeroth law of thermodynamics. When two systems are each in thermal equilibrium with a third system, the first two systems are in thermal equilibrium with each other. This property makes it meaningful to use thermometers as the “third system” and to define a temperature scale.

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41. Write first law of thermodynamics and its mathematical equation?

Answer: The first law of thermodynamics, or the law of conservation of energy. The change in a system's internal energy is equal to the difference between heat added to the system from its surroundings and work done by the system on its surroundings.

The equation for the first law of thermodynamics is given as;

ΔU = q + W

Where,

ΔU = change in internal energy of the system.

q = algebraic sum of heat transfer between system and surroundings.

W = work interaction of the system with its surroundings.

42.What are the characteristics of good qualities of energy (3)

Answer: Characteristics of a good source of energy are:

It should be a sustainable and renewable source of energy.
It should have a high calorific value.
It should be easily accessible and provide energy for the maximum period of time.
It should not cause pollution.

43.Write examples of diffusions?

Answer: Examples of Diffusion:

- A tea bag immersed in a cup of hot water will diffuse into the water and change its colour.
- A spray of perfume or room freshener will get diffused into the air by which we can sense the odour.
- Sugar gets dissolved evenly and sweetens the water without having to stir it.

44.Explain principle of mass spectrometry? (5)

Answer: Mass spectrometry is a powerful analytical technique used to quantify known materials, to identify unknown compounds within a sample, and to elucidate the structure and chemical properties of different molecules.

The basic principle of mass spectrometry (MS) is to generate ions from either inorganic or organic compounds by any suitable method, to separate these ions by their mass-to-charge ratio (m/z) and to detect them qualitatively and quantitatively by their respective m/z and abundance.

The first step in the mass spectrometric analysis of compounds is the production of gas phase ions of the compound, basically by electron ionization. This molecular ion undergoes

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fragmentation. Each primary product ion derived from the molecular ion, in turn, undergoes fragmentation, and so on. The ions are separated in the mass spectrometer according to their mass-to-charge ratio, and are detected in proportion to their abundance. A mass spectrum of the molecule is thus produced. It displays the result in the form of a plot of ion abundance versus mass-to-charge ratio. Ions provide information concerning the nature and the structure of their precursor molecule. In the spectrum of a pure compound, the molecular ion, if present, appears at the highest value of m/z (followed by ions containing heavier isotopes) and gives the molecular mass of the compound.

45. Write a note on Metabolic Pathway? (5)

Answer: A metabolic pathway is a linked series of chemical reactions occurring within a cell. The reactants, products, and intermediates of an enzymatic reaction are known as metabolites, which are modified by a sequence of chemical reactions catalyzed by enzymes. In most cases of a metabolic pathway, the product of one enzyme acts as the substrate for the next. However, side products are considered waste and removed from the cell. These enzymes often require dietary minerals, vitamins, and other cofactors to function

Different metabolic pathways function based on the position within a eukaryotic cell and the significance of the pathway in the given compartment of the cell. For instance, the, electron transport chain and oxidative phosphorylation all take place in the mitochondrial membrane. In contrast, glycolysis, pentose phosphate pathway, and fatty acid biosynthesis all occur in the cytosol of a cell.

There are two types of metabolic pathways that are characterized by their ability to either synthesize molecules with the utilization of energy (anabolic pathway), or break down complex molecules and release energy in the process (catabolic pathway).

46.What are the types of RNA? (5)

Answer: **Messenger RNA (mRNA):**

Messenger RNA (mRNA) carries the genetic code from DNA in a form that can be recognized to make proteins. The coding sequence of the mRNA determines the amino acid sequence in the protein produced. Once transcribed from DNA, eukaryotic mRNA briefly exists in a form called “precursor mRNA (pre-mRNA)” before it is fully processed into mature mRNA.

Ribosomal RNA (rRNA):

Ribosomal RNA is the catalytic component of ribosomes. In the cytoplasm, rRNAs and protein components combine to form a nucleoprotein complex called the ribosome which binds mRNA and synthesizes proteins (also called translation).

Transfer RNA (tRNA):

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Transfer RNA is a small RNA chain of about 80 nucleotides. During translation, tRNA transfers specific amino acids that correspond to the mRNA sequence into the growing polypeptide chain at the ribosome.

47.Second law of thermodynamics (deep analysis)?

Answer: Second law of thermodynamics, statement describing the amount of useful work that can be done from a process that exchanges or transfers heat.

A cyclic transformation whose only final result is to transfer heat from a body at a given temperature to a body at a higher temperature is impossible.

The net effect would be a flow of heat from a lower temperature to a higher temperature, thereby violating the second (Clausius) form of the second law. Conversely, if the second form were possible, then the heat transferred to the higher temperature could be used to run a heat engine that would convert part of the heat into work. The final result would be a conversion of heat into work at constant temperature—a violation of the first (Kelvin) form of the second law.

48.Lattice enthalpy (5)

Answer: Lattice enthalpy is a measure of the strength of the forces between the ions in an ionic solid. The greater the lattice enthalpy, the stronger the forces. Those forces are only completely broken when the ions are present as gaseous ions, scattered so far apart that there is negligible attraction between them.

The two main factors affecting lattice enthalpy are

- The charges on the ions and
- The ionic radii (which affects the distance between the ions).

Calculating Lattice Enthalpy

It is impossible to measure the enthalpy change starting from a solid crystal and converting it into its scattered gaseous ions. It is even more difficult to imagine how you could do the reverse - start with scattered gaseous ions and measure the enthalpy change when these convert to a solid crystal. Instead, lattice enthalpies always have to be calculated, and there are two entirely different ways in which this can be done.

- You can use a Hess's Law cycle (in this case called a Born-Haber cycle) involving enthalpy changes which can be measured. Lattice enthalpies calculated in this way are described as experimental values.
- Or you can do physics-style calculations working out how much energy would be released, for example, when ions considered as point charges come together to make a lattice. These are described as theoretical values. In fact, in this case, what you are actually calculating are properly described as lattice energies.

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49.Ion exchange chromatography?

Answer: Ion-exchange chromatography (IEC) is part of ion chromatography which is an important analytical technique for the separation and determination of ionic compounds, together with ion-partition/interaction and ion-exclusion chromatography.

Ion chromatography separation is based on ionic (or electrostatic) interactions between ionic and polar analytes, ions present in the eluent and ionic functional groups fixed to the chromatographic support. **Two distinct mechanisms as follows**; ion exchange due to competitive ionic binding (attraction) and ion exclusion due to repulsion between similarly charged analyte ions and the ions fixed on the chromatographic support, play a role in the separation in ion chromatography.

50.X-ray crystallography?

Answer: X-ray crystallography is the experimental science determining the atomic and molecular structure of a crystal, in which the crystalline structure causes a beam of incident X-rays to diffract into many specific directions. By measuring the angles and intensities of these diffracted beams, a crystallographer can produce a three-dimensional picture of the density of electrons within the crystal. From this electron density, the mean positions of the atoms in the crystal can be determined, as well as their chemical bonds, their crystallographic disorder, and various other information.

51.Ion exchange chromatography procedure (5)

Answer: Key steps in the ion exchange chromatography procedure are listed below:

- An impure protein sample is loaded into the ion exchange chromatography column at a particular pH.
- Charged proteins will bind to the oppositely charged functional groups in the resin
- A salt gradient is used to elute separated proteins. At low salt concentrations, proteins having few charged groups are eluted and at higher salt concentrations, proteins with several charged groups are eluted.
- Unwanted proteins and impurities are removed by washing the column.

52.Write the instruments of HLPC? (5)

Answer: **HPLC Mobile Phase**

Mobile phase serves to transport the sample to the system. Essential criteria of mobile phase are inertness to the sample components. Pure solvents or buffer combinations are commonly used. The mobile phase should be free of particulate impurities and degassed before use.

HPLC Mobile Phase Reservoirs

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These are inert containers for mobile phase storage and transport. Generally transparent glass bottles are used so as to facilitate visual inspection of mobile phase level inside the container. Stainless steel particulate filters are provided inside for removal of particulate impurities in the mobile phase if any.

HPLC Pumps

Variations in flow rates of the mobile phase effect elution time of sample components and result in errors. Pumps provide constant flow of mobile phase to the column under constant pressure.

HPLC Injectors

Injectors are used to provide constant volume injection of sample into the mobile phase stream. Inertness and reproducibility of injection are necessary to maintain high level of accuracy.

HPLC Column

A column is a stainless steel tube packed with stationary phase. It is a vital component and should be maintained properly as per supplier instructions for getting reproducibility separation efficiency run after run.

Column Oven

Variation of temperature during the analytical run can result in changes of retention time of the separated eluting components. A column oven maintains constant column temperature using air circulation. This ensures a constant flow rate of the mobile phase through the column

Detector

A detector gives specific response for the components separated by the column and also provides the required sensitivity. It has to be independent of any changes in mobile phase composition. Majority of the applications require UV-VIS detection though detectors based on other detection technique are also popular these days.

Data Acquisition & Control

Modern HPLC systems are computer based and software controls operational parameters such as mobile phase composition, temperature, flow rate, injection volume and sequence and also acquisition and treatment of output.

These are the main parts of a basic HPLC system more specialized equipment might also have solvent selection valves, vacuum degasser, auto samplers, column switchers, pre or post column derivatization and fraction collectors. These are all covered at length in our upcoming online certificate program on HPLC.

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